

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
3rd Semester of B. Pharm. Examination
University Theory Examination April/May 2016
PH204 Pharmaceutical Analysis I

Date: 12.04.2016, Tuesday

Time: 1:30 p.m to 4:30 p.m

Maximum Marks: 80

Instructions:

- 1. Figures to right indicate marks.**
- 2. Section wise separate answer book to be used.**
- 3. Draw neat sketches wherever necessary.**

SECTION- I

Q 1 Attempt any **TWO** of the following

- | | | |
|---|--|----|
| A | What is hydrolysis. Derive equation for finding pH for Salt of Strong base and Weak Acid | 10 |
| B | 1. Explain common ion effect and Law of mass action in detail. | 05 |
| | 2. Define Buffer. Derive Henderson-Hasselbach equation. | 05 |
| C | 1. What is error? Classify the error and explain error minimization techniques. | 05 |
| | 2. Enlist analytical method validation parameters. Explain Accuracy and Precision in detail. | 05 |

Q 2 Attempt any **TWO** of the following

- | | | |
|---|---|----|
| A | 1. What is QA and QC. Give functions of QC. | 05 |
| | 2. Explain effect of pH in extractability of drugs. | 05 |
| B | Explain the steps involved in gravimetric analysis. | 10 |
| C | Give brief account of solubility product, Precipitation techniques and pharmacopoeial applications of Gravimetric analysis. | 10 |

SECTION- II

Q 3 Attempt any **TWO** of the following

- | | | |
|---|--|---|
| A | (1) What is Karl Fisher Titration? Give the significance of each reagent in Karl fisher titration. | 4 |
| | (2) Define a) Oxidation, | 4 |
| | b) Redox Potential, | |

c) Oxidation Number

d) Self Indicator

(3) Calculate the Redox Potential of the following cell 2



$$E^0_{\text{Zn}|\text{Zn}^{2+}} = -0.76\text{V} \quad \text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$$

$$E^0_{\text{Cu}^{2+}|\text{Cu}} = 0.34\text{V} \quad \text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$$

B Explain Iodometry and iodimetry, and state the difference between them. 10

C (1) Discuss the characteristic features of EDTA as a complexometric agent. 4

(2) Explain any three different types of complexometric titrations 6

Q 4 Attempt any TWO of the following

A (1) When are non-aqueous titrations preferred over aqueous titration method. 3

(2) Explain the leveling and differentiating effects seen in non-aqueous titration. 7

B (1) Explain in brief Gasometry 3

(2) Explain Fajjan's Method of Precipitation titration with the chemical reactions involved. 5

(3) Name the indicator used in Fajan's titration and Mohrs titration method. 2

C Explain Volhards method of Precipitation titration. 10
